

DineAI Framework - Professional Development Roadmap

This roadmap summarizes the remaining development phases required to transition the completed DineAI framework into a production-ready AI restaurant recommendation system.

Phase 1 – Dataset Pipeline

Build an orchestration pipeline that converts `recipes.csv` into standardized artifacts by executing Schema Mapping, Validation, Feature Engineering, Text Builder, Embedding Builder, and FAISS Builder.

Phase 2 – Dataset Manager

Automatically detect datasets, validate processed artifacts, rebuild when necessary, and load restaurants without requiring manual preprocessing.

Phase 3 – Production Dataset

Prepare a real restaurant dataset and generate `recipes.pkl`, `embeddings.npy`, `faiss.index`, and `metadata.json`.

Phase 4 – Real LLM Integration

Replace the Mock Provider with a real Qwen provider while keeping the provider architecture extensible.

Phase 5 – Functional Validation

Execute realistic restaurant queries and verify semantic retrieval, filtering, ranking, prompting, and final response quality.

Phase 6 – FastAPI

Expose the application through REST endpoints such as `/chat`, `/health`, `/reload`, and `/restaurant`.

Phase 7 – Docker

Containerize the application for reproducible deployment.

Phase 8 – AWS Deployment

Deploy the application on AWS using the generated datasets and production configuration.

Recommended Execution Flow

recipes.csv → Dataset Pipeline → recipes.pkl + embeddings.npy + faiss.index +
metadata.json → Runtime Pipeline → Semantic Search → Filtering → Ranking → Prompt
Builder → Qwen → Final Response